BAYBASI PIXEL WALL — COLUMN CONTROLLER rev A

#FLG0
+5V
PWR_FLAG

#FLG1
GND
PWR_FLAG

POWER: reverse-protected, LC-filtered 5 V from this column's LED supply

3V3E: dedicated rail, W5500 only

J13
VIN_RAW
GND
5V IN

V5_PROT
VGATE
R13
100k
Q1
VGATE_TP06
VIN_RAW

V5_PROT
L1
4.5uH 2A

+5V
C1
100uF 16V
GND

+5V
C3
100nF
GND

U3
5V
VI
VO
GND
AM1117-3.3
+3V3E
GND

+3V3E
C4
100nF
GND

+3V3E
C2
100uF
GND

GND
+3V3E
MOSI
SCK
CS
INT
W5500lo J1

+3V3E
C7
100nF
GND

+5V
C5
100nF
GND

+5V
C6
100nF
GND

GND
+3V3E
RST
MISO
W5500lo J2

ETH_RST
R14
10k
+3V3E

+5V
CHIP_PU
5V
3V3
GPIO43/U0TXD
GPIO44/U0RXD
GPIO15/ADC2_CH4/32K_P
GPIO16/ADC2_CH5/32K_N
GPIO20/USB_D+
GPIO19/USB_D-
GPIO39/MTCK
GPIO40/MTDO
GPIO41/MTDI
GPIO42/MTMS
GPIO0
GPIO1/ADC1_CH0
GPIO2/ADC1_CH1
GPIO3/ADC1_CH2
GPIO4/ADC1_CH3
GPIO5/ADC1_CH4
GPIO6/ADC1_CH5
GPIO7/ADC1_CH6
A1
GPIO8/ADC1_CH7
GPIO9/ADC1_CH8
GPIO10/ADC1_CH9
GPIO11/ADC2_CH0
GPIO12/ADC2_CH1
GPIO13/ADC2_CH2
GPIO14/ADC2_CH3
GPIO17/ADC2_CH6
GPIO18/ADC2_CH7
GPIO21
GPIO35
GPIO36
GPIO37
GPIO38
GPIO45
GPIO46
GPIO47
GPIO48
ESP32-S3-DevKitC
GND
D6_3V3
D4_3V3
D5_3V3
D1_3V3
D2_3V3
D3_3V3
SYNC_IN
ETH_CS
ETH_MOSI
ETH_SCK
ETH_MISO
ETH_INT
D7_3V3
D12_3V3

J14
SYNC
GND
SYNC IN (opt)

LEVEL SHIFT 3V3 -> 5V (AHCT: TTL thresholds)

+5V
U1
SN74AHCT245PWR
D1_3V3
D2_3V3
D3_3V3
D4_3V3
D5_3V3
D6_3V3
GND
+5V
A->B
CE
VCC
GND
B0
B1
B2
B3
B4
B5
B6
B7
D1_5V
D2_5V
D3_5V
D4_5V
D5_5V
D6_5V

+5V
U2
SN74AHCT245PWR
D7_3V3
D8_3V3
D9_3V3
D10_3V3
D11_3V3
D12_3V3
GND
+5V
A->B
CE
VCC
GND
B0
B1
B2
B3
B4
B5
B6
B7
D7_5V
D8_5V
D9_5V
D10_5V
D11_5V
D12_5V

12 x SERIES-TERMINATED OUTPUT DATA + GND, no +5 V pin

D1_5V
R1
330
J1
OUT
GND
PANEL 1

D2_5V
R2
330
J2
OUT
GND
PANEL 2

D3_5V
R3
330
J3
OUT
GND
PANEL 3

D4_5V
R4
330
J4
OUT
GND
PANEL 4

D5_5V
R5
330
J5
OUT
GND
PANEL 5

D6_5V
R6
330
J6
OUT
GND
PANEL 6

D7_5V
R7
330
J7
OUT
GND
PANEL 7

D8_5V
R8
330
J8
OUT
GND
PANEL 8

D9_5V
R9
330
J9
OUT
GND
PANEL 9

D10_5V
R10
330
J10
OUT
GND
PANEL 10

D11_5V
R11
330
J11
OUT
GND
PANEL 11

D12_5V
R12
330
J12
OUT
GND
PANEL 12

Connectivity is by net label; labels sit on pin endpoints.

U1/U2 pins 8,9 (A6,A7) tied low: unused AHCT inputs must not float.

+5V
C9
100nF
GND

+5V
C8
100nF
GND

Sheet: /	
File: baybasi-ctrl.kicad_sch	
Title:	
Size: A2	Date:
KiCad E.D.A. 10.0.6	**Rev:**
	Id: 1/1